High-density livestock operations, crop field application of manure, and risk of community-associated methicillin-resistant Staphylococcus aureus infection, Pennsylvania, USA

Context Nearly 80% of antibiotics in the United States are sold for use in livestock feeds. The manure produced by these livestock contains antibiotic-resistant bacteria, resistance genes, and antibiotics, and is subsequently applied to crop fields where it may put community members at risk for antibiotic-resistant infections. Objective To assess the association between individual exposure to swine and dairy/veal industrial agriculture and risk of methicillin-resistant Staphylococcus aureus (MRSA) infection. Design, Setting, and Participants A population-based, nested case-control study of Geisinger primary care patients in Pennsylvania from 2005–2010. Incident MRSA cases were identified using electronic health records, classified as community-associated or healthcare-associated, and frequency-matched to randomly selected controls and patients with skin and soft tissue infection. Nutrient management plans were used to create two exposure variables: seasonal crop field manure application and number of livestock at the operation. In a sub-study we collected 200 isolates from patients stratified by location of diagnosis and proximity to livestock operations. Main outcome measures Community-associated MRSA, healthcare associated-MRSA, and skin and soft tissue infection status (with no history of MRSA) compared to controls. Results From 446,480 patients, 1539 community-associated MRSA, 1335 healthcare-associated MRSA, 2895 skin and soft tissue infection cases, and 2914 controls were included. After adjustment for MRSA risk factors, the highest quartile of swine crop field exposure was significantly associated with community-associated MRSA, healthcare-associated MRSA, and skin and soft tissue infection case status (adjusted odds ratio, 1.38 [95% CI, 1.13–1.69], 1.30 [95% CI, 1.05–1.61], and 1.37 [95% CI, 1.18–1.60], respectively); and there was a trend of increasing odds across quartiles for each outcome (all P for trend $\leq$0.01). There were similar but weaker associations of swine operations with community-associated MRSA and skin and soft tissue infection. Molecular testing of 200 isolates identified 31 unique spa types, none of which corresponded to CC398, but some have been previously found in swine. Conclusion Proximity to swine manure application to crop fields and livestock operations each was associated with MRSA and skin and soft tissue infection. These findings contribute to the growing concern about the potential public health impacts of high-density livestock production.